

Cheat Sheet: Hardware & IO Allocation (DCS 1)

Exam: 20 questions · 25 min · Moodle · week 39 · materials allowed · 20 p.

1. DCS in general

- **DCS** = Distributed Control System
- PLC and DCS → **same core task**: read inputs → run program → write outputs
- **DCS characteristics** (≠ PLC):
 - plant-wide, **one** integrated system
 - functions **distributed across specialised stations**
 - **continuous operator interaction** (24/7, alarms, trends, setpoints)
 - scalable (add stations/nodes/subsystems)
 - long lifecycle, gradual modernisation
 - central engineering (EAS)
- **PLC** → machines, small I/O count, often autonomous, 1 to few controllers
- **DCS** → power plant, pulp mill, refinery, water process, many I/Os
- PLC CPU ≈ **process station** · TIA project ≈ **EAS** · HMI/SCADA ≈ **operator station**
- History: Damatic 1979 → XD 1988 → metsoDNA 2000 → **Valmet DNA 2015**
- Others: Siemens PCS 7 · ABB 800xA · Honeywell Experion · Emerson DeltaV

2. Stations & roles

Station	Code	OS	Keywords
Process Control Station	P / PCS	Linux	real-time control, FbCAD logic, SeqCAD, fieldbuses via FBC
Backup Station	B / BU	Windows	not a real backup! distributes software · holds all node applications · nodes load after boot · auto restart after power failure · versions
Alarm Station	A / ALS	Windows	collect alarms · time stamp · prioritise · distribute to OPS
Engineering Activity Server	E / EAS	Windows	engineering DB · FbCAD · Function Test · versions · multi-user
Operator Station	O / OPS	Windows	monitoring and operation (DNA Operate)
Diagnostics	D / DIA	Windows	diagnostics
OPC station	O	Windows	connection to other systems (GO01)

- Network: **redundant Ethernet** · Valmet protocol over **UDP/IP**
- **ACN** = Application and Control Node (hardware)
 - **ACN MR** (Modular Rail) → small → **GP01** (lab DP66)
 - **ACN CS** → industrial PC → EP01/EP02
 - **ACN RT** → most powerful option
- JAMK: servers **virtualised** · GB01 + G109 + G1A1 on **one** VM

3. Naming policy

Subsystem + (control room) + type + number

- **G** = automation lab DP66 · **E** = energy lab
- Type: **P** process · **B** backup · **A** alarm · **E** EAS · **O** operator/OPC · **D** diagnostics

Name	Meaning
GP01	G · process station · 01
GB01	G · backup station · 01
G1A1	G · control room 1 · alarm · 1
G109	G · control room 1 · operator · 9
EB01	E · backup · 01
EAS-G	engineering server of G
GO01	G · OPC server · 01

Slots

- then: real cards (CPU, NCU, FBC, BU) · now: **virtual** (Slot Controller)
- GP01: **slot 2 = ACN IO · 4, 5, 6 = Profibus DP · 9 = PROFINET**
- NCU2 = network communication

4. Hardware & system layout

- **GP01** = ACN MR + local **ACN IO** (M80) in the I/O cabinet, DP66
- Server VMs in the server room: **EAS-G**, EAS-E, **GB01/G109/G1A1**, **GO01**, SIMU, E-INFO
- Network: VLAN121 · 192.168.45.0/24 · GP01 = .62
- Access: remote.jamk.fi → **EAS-G** · **sign out** afterwards (licence!)

Terms

- **FBC** = Field Bus Controller (fieldbus interface of the PCS)
- **IBC** = I/O bus controller (I/O group 0-15)
- **IPSP** = power supply · **MB8** = mounting base for 8 units
- max. **16 cards** per I/O unit

Card layout GP01 (IBC 0)

Place	0	1	2	3	4	5	6	7
Card	DI8P	DI8P	DO8P	DO8P	AI8C	AO4C	AO4C	free

I/O cards

Card	Channels	Keywords
DI8	8	digital input 24 V · PNP/NPN · 1 > 4.5 mA · 0 < 2 mA
DO8P	8	digital output · relay · < 200 mA · safe state configurable
AI8C/V	8	0/4-20 mA or 0/2-10 V · own 24 V supply (C) · wire break/short circuit
AO4C/V	4	0/4-20 mA or 0/2-10 V · wire break/short circuit
TI4	4	Pt100 · 3-/4-wire · -50...+400 °C

Wiring (path of the signal)

Field device → device cable (2-pair) → **field box** → **trunk cable** (24-pair) → **cross connection** in the I/O cabinet → **CXW/CXS/CXR** → I/O card

HW engineering: order

1. Initial data (PI diagram)
2. Select field devices
3. **Loop list + IO list**
4. Loop diagram (cabling, cross connection)

5. IO address

FBC : IBC : card : channel → everything counted **from 0** (FBC from 2)

- 2:0:0:0 → DI8, channel 0 (switch HS)
- 2:0:2:0 → DO8, channel 0 (red light) · 2:0:2:1 → yellow light
- 2:0:4:0 → AI8, channel 0 (temperature TT)
- Parameters in the I/O module: **FBC** (2-15) · **IBC** (0-15) · **card** (0-15) · **channel** (0-7)
- AI additionally: min/max · range · filter · line fault limits

IO list — columns

station · FBC · IBC · card type · place · channel · **device tag** · **position** · signal · operation

- **Device tag** = device in the field (LT-1, LS-1)
- **Position** = function in the system (LI-1, LZ-1)
- both taken from the **PI diagram**
- procedure: count signals → calculate cards → assign channels

6. FbCAD & DNA Explorer

- **FbCAD** = graphical programming (function blocks)
- **DNA Explorer** = manage applications in the EAS database

Administration part

Attribute	Remember
Name	pr:TAG.F · starts with pr:
Tag	TAG · unique ID in the database
Package	station it runs on → GP01
Execution / Order	cycle time (1000 ms) / order (20)
Loop name	comment only

- **Same tag → the new loop overwrites the old one!**
- **Direct Access Port** = **global** variable → name **unique system-wide** → link to the control room
- **Interfaces out** = output other loops can read · unique only within the diagram
- Input modules **.I** on the left · output modules **.O** on the right
- Change values: **Edit** → **Values** (never double-click!)
- Blocks: **am** analog measurement · **mtr** motor · **mgv** magnetic valve · **pid** controller · **ccoX** conditional copy

Wire colours (Draw → Scalar **autocolor** wire)

Colour	Type
Black	bin (binary)
Green	ana (analog)
Yellow	float
Red	fails (fault bits)
Violet	bo (binary output)
Blue / light blue	intl / ints
Grey	general

7. Download

1. Save to the **workspace**
2. **File** → **Check** → until error-free
3. Move to the **repository**

4. **Check Function + IO address check**
 5. Mode **Real** (not Virtual)
 6. **Download to...** → Add/Replace → **Accept to backup** → OK
 7. Only download **in the lab!**
- **Real** → only from repository · checks **IO addresses**
 - **Virtual** → workspace or repository · **without** IO addresses

8. Function Test tool

- Start: **File** → **Test** (FbCAD) or Test Function (Explorer)
- download to the PCS first
- **live values** from the real/virtual runtime environment
- set **test points** (blue values)
- **change values & parameters**
- **history** + trends (step back in time)
- test points **not saved** · diagram stays **unchanged**

Quick test

Question	Answer
DCS?	Distributed Control System
DCS vs. PLC?	distributed · plant-wide · operator 24/7 · scalable
GB01?	G · backup station · 01
Backup station?	distributes software/applications · auto restart · not a real backup
Black wire?	bin
Same tag?	overwrites the old loop
Function Test?	test live · change values · history
PCS OS?	Linux
ACN IO on GP01?	FBC 2
2:0:4:0?	FBC 2 · IBC 0 · card 4 · channel 0
Channels DI8/DO8/AI8/AO4/TI4?	8/8/8/4/4
Cards per I/O unit?	16
Real download from?	Repository

Pitfalls: O = operator **and** OPC · backup ≠ data backup · virtual checks no I/Os · counting starts at 0